Imperial College London

02/13/2023

Fair and skill-diverse student group formation via constrained k-way graph partitioning

AAAI 2023 – Artificial Intelligence for Education

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Group formation in education

1. Students often learn best in small groups.
2. Group formation is a **manual**, **subjective** and **laborious** task. As a result, students are often assigned at random, which leads to unequal student experiences.
3. **Diversifying skill sets** within groups has shown to create a stimulating and productive environment for learning [3, 4].
4. But group formation often has to satisfy **multiple constraints**, e.g. diversity and inclusion, and group sizes.

We propose...

1. To learn student skill-sets from only course mark data in an unsupervised manner using the graph Laplacian.
2. An algorithm for fair and skill-diverse student group formation.

The effectiveness of our algorithm is demonstrated on undergraduate course marks from $N = 54$ students in $L = 23$ courses at Imperial College London.

Standard analysis of course marks

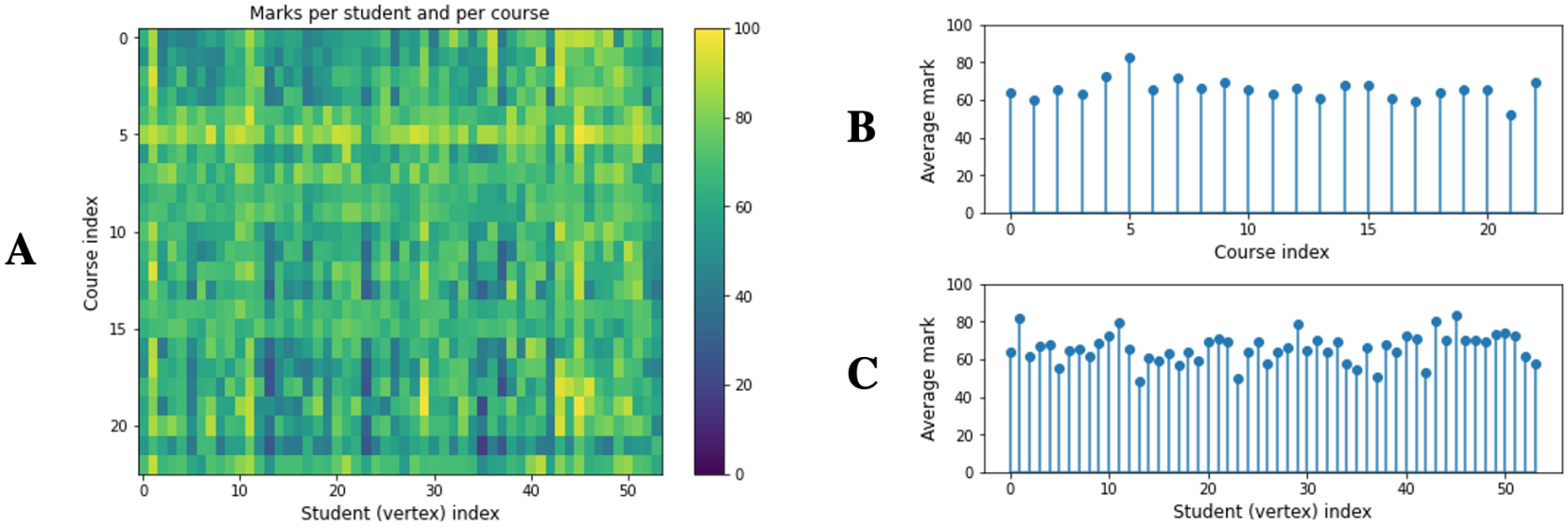

Figure 1: The course marks of $N = 54$ EIE students in $L = 23$ courses viewed as A) marks per student and per course, B) average mark per course, or C) average mark per student. [5]

The n-th student resides in an L-dimensional vector space, with position given by their feature vector of course marks, $\mathbf{r}_n \in \mathbb{R}^L$, here $L = 23$ (# of courses).

Learning skill-sets via manifold learning

The Laplacian eigenmap

- Define a graph using a weighted adjacency matrix, $\mathbf{W} \in \mathbb{R}^{N \times N}$, where each vertex is a student. Compute the elements of $\mathbf{W}$ as

$$W_{mn} = \begin{cases} \exp\left(\frac{Corr(\mathbf{r}_m, \mathbf{r}_n)^2}{A}\right) & \text{if} \quad Corr(\mathbf{r}_m, \mathbf{r}_n) \geq B, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

- Define the graph Laplacian, $\mathbf{L} = \mathbf{D} - \mathbf{W}$, where $\mathbf{D} \in \mathbb{R}^{N \times N}$ is a diagonal matrix with elements representing the degree of each vertex.
- An eigen-decomposition of the graph Laplacian, $\mathbf{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$, yields the matrix of eigenvectors $\mathbf{U} \in \mathbb{R}^{N \times N}$ and eigenvalues $\mathbf{\Lambda} \in \mathbb{R}^{N \times N}$.

Each student / vertex is represented in a new M-dimensional space ($M < L$) using the spectral vector, $\mathbf{q}_n = [u_1(n), ..., u_{M-1}(n)]$.

Spectral vector encodes student skill-sets

A Laplacian eigenmap is constructed, whereby the position of the n-th student is represented using an 3-dimensional spectral vector, $\mathbf{q}_n = [u_1(n), u_2(n), u_3(n)]$.

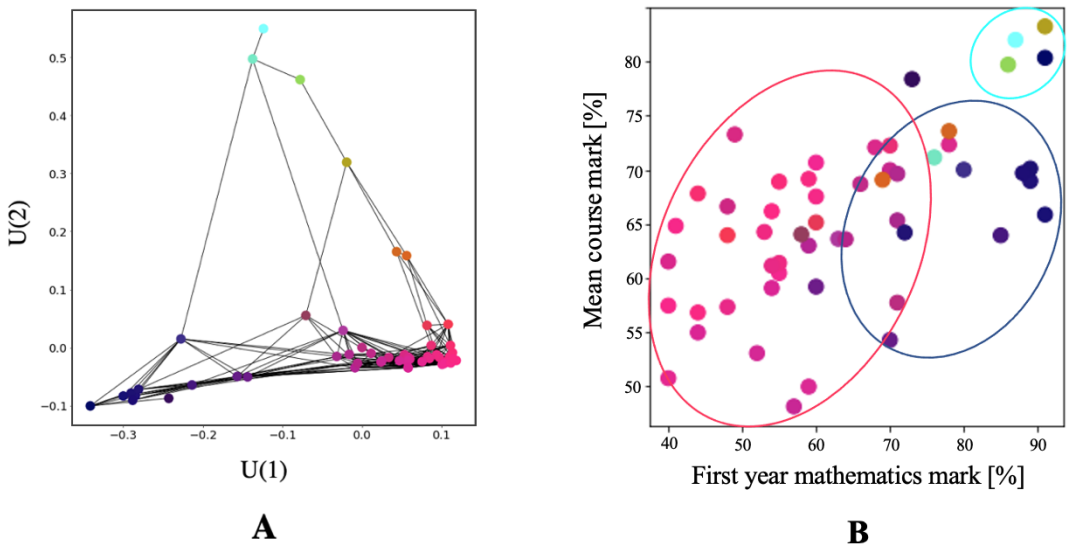

Figure 2: Dimensionality reduction of course marks and interpretation. A) Students visualised using the first two components of their 3D spectral vector. B) Students cluster into three skill-sets: below average in mathematics (pink ellipse), above average in mathematics (dark blue ellipse), and consistent high achievers (teal ellipse).

Fair and skill-diverse group formation - I

Students with similar skill-sets are closer together in the Laplacian eigenmap, i.e. they have similar spectral vectors, $\mathbf{q}_n$.

Our definition of a skill-diverse group

- A group made up of students with **maximally different spectral vectors**.

The similarity between spectral vectors of students m and n is quantified with the Euclidean distance,

$$d_{mn} = \|\mathbf{q}_m - \mathbf{q}_n\|_2. \tag{2}$$

Fair and skill-diverse group formation - II

Fairness, e.g. for diversity and inclusion, is a common constraint in group formation. One fairness constraint is the *balance* [2, 1], B_{cs}, of a sensitive attribute, s, in a group, c, given by,

$$B_{cs} = \min\left\{R_{cs}, \frac{1}{R_{cs}}\right\}, \tag{3}$$

where $R_{cs} = \frac{a_{cs}}{a_s}$, a_{cs} is the ratio of sensitive attribute s in the group c, and a_s is the ratio of sensitive attribute s in the population.

Our definition of a fair group

- A group that satisfies a lower-bound constraint, B_{L_s}, on the balance of each sensitive attribute, B_{cs}.

Fair and skill-diverse group formation - III

Graph partitioning clusters graph vertices into mutually exclusive groups. A k-way graph partition is defined as the division of graph vertices into k disjoint groups.

The graph partition vector, $\mathbf{w} \in \{0, 1\}^{|E|}$, indicates which edges are connected (value 1) and disconnected (value 0) in the resulting k-way graph partition.

Two vertices that remain connected, $w_{mn} = 1$, designate two students that will be grouped together.

Optimisation problem

We formulate the fair and skill-diverse student group formation problem as a constrained integer programming problem, given by

$$\max_{\{w_{mn}\}_{\forall\{m,n\}\in E}} \sum_{\{m,n\}\in E} d_{mn}\, w_{mn}$$

Promotes skill diversity

$w_{mn} = 1$ if in same group

$d_{mn} = \|\mathbf{q}_m - \mathbf{q}_n\|_2$

Optimisation problem

Fair and skill-diverse student group formation is possible via the following constrained integer programming problem,

$$\max_{\{w_{mn}\}_{\forall\{m,n\}\in E}} \sum_{\{m,n\}\in E} d_{mn} w_{mn}$$

subject to

$$w_{mn} + w_{mo} - w_{no} \leq 1 \qquad \forall m, n, o \in V : m \neq n \neq o \,.$$

Triangle inequalities

All pairs of three different students

Optimisation problem

We formulate the fair and skill-diverse student group formation problem as a constrained integer programming problem, given by

$$\max_{\{w_{mn}\}_{\forall\{m,n\}\in E}} \sum_{\{m,n\}\in E} d_{mn} w_{mn}$$

subject to

$$w_{mn} + w_{mo} - w_{no} \leq 1 \quad \forall m, n, o \in V : m \neq n \neq o$$

$$|\mathbf{w}(\delta(m))| + 1 \geq F_L \quad \forall m \in V$$

Lower bound of group size

Indices of edges adjacent to vertex m

Optimisation problem

We formulate the fair and skill-diverse student group formation problem as a constrained integer programming problem, given by

$$\max_{\{w_{mn}\}_{\forall\{m,n\}\in E}} \sum_{\{m,n\}\in E} d_{mn} w_{mn}$$

subject to

$$w_{mn} + w_{mo} - w_{no} \leq 1 \quad \forall m, n, o \in V : m \neq n \neq o$$

$$|\mathbf{w}(\delta(m))| + 1 \geq F_L \quad \forall m \in V$$

$$|\mathbf{w}(\delta(m))| + 1 \leq F_U \quad \forall m \in V$$

Upper bound of group size

Number of vertices in same group as m

Optimisation problem

We formulate the fair and skill-diverse student group formation problem as a constrained integer programming problem, given by

$$\max_{\{w_{mn}\}_{\forall\{m,n\}\in E}} \sum_{\{m,n\}\in E} d_{mn} w_{mn}$$

subject to

$$w_{mn} + w_{mo} - w_{no} \leq 1 \quad \forall m, n, o \in V : m \neq n \neq o$$

$$|\mathbf{w}(\delta(m))| + 1 \geq F_L \quad \forall m \in V$$

$$|\mathbf{w}(\delta(m))| + 1 \leq F_U \quad \forall m \in V$$

$$w_{mn} \in \{0, 1\} \quad \forall \{m, n\} \in E$$

Constrained to integer value

Optimisation problem

We formulate the fair and skill-diverse student group formation problem as a constrained integer programming problem, given by

$$\max_{\{w_{mn}\}_{\forall\{m,n\}\in E}} \sum_{\{m,n\}\in E} d_{mn} w_{mn}$$

subject to

$$\begin{aligned}
&w_{mn} + w_{mo} - w_{no} \leq 1 \quad \forall m, n, o \in V : m \neq n \neq o \\
&|\mathbf{w}(\delta(m))| + 1 \geq F_L \quad \forall m \in V \\
&|\mathbf{w}(\delta(m))| + 1 \leq F_U \quad \forall m \in V \\
&w_{mn} \in \{0, 1\} \quad \forall \{m, n\} \in E \\
&B_{L_s} \leq B_{cs}(m) \quad \forall s \in S, \forall m \in V.
\end{aligned}$$

Lower bound for balance of s

Balance of s in group containing m

Set of all sensitive attributes

Real-world case study

Ten students were chosen at random from the dataset to test our algorithm. Group sizes were constrained to $F_L = F_U = 5$.

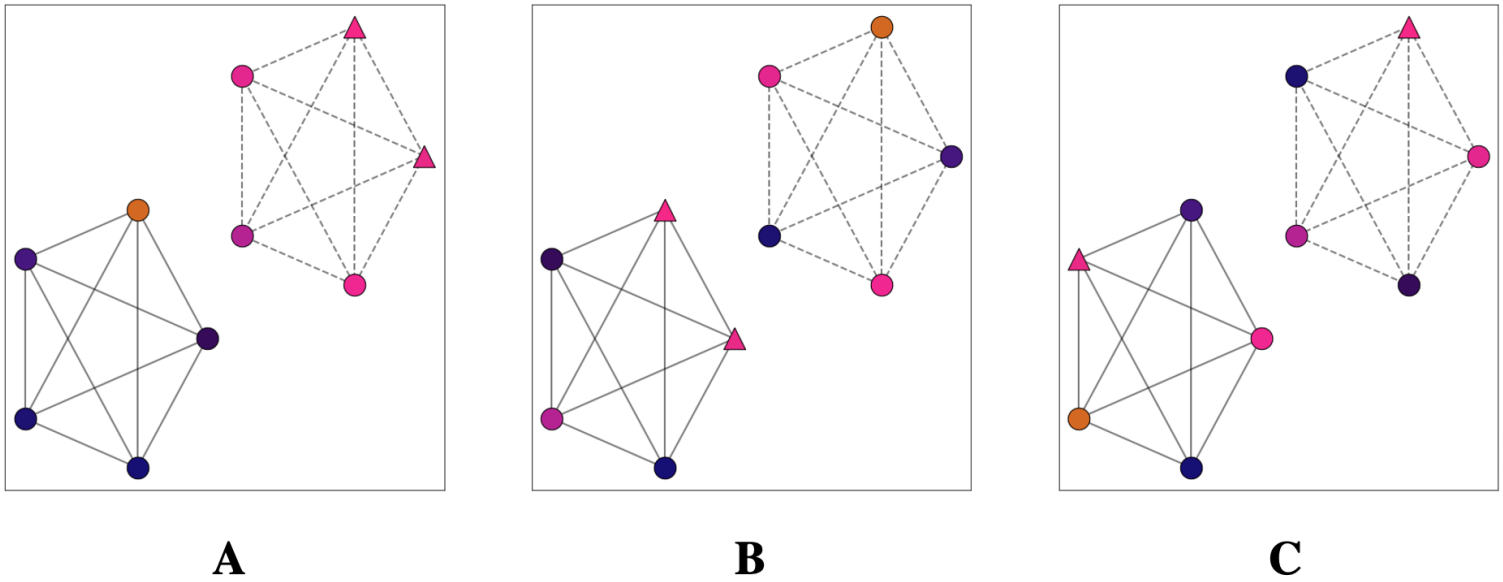

Figure 3: Groups formed using different optimisation objectives. A) Minimising the objective gives groups of students with similar skill-sets. B) Maximising the objective gives skill-diverse groups of students. C) Maximising the objective and introducing a lower bound on the balance, $B_{L_s} = 1$, of a sensitive attribute gives skill-diverse and fair groups of students. Students with a sensitive attribute are denoted with a triangle.

Real-world case study

Investigation of the in-group correlation between student course marks.

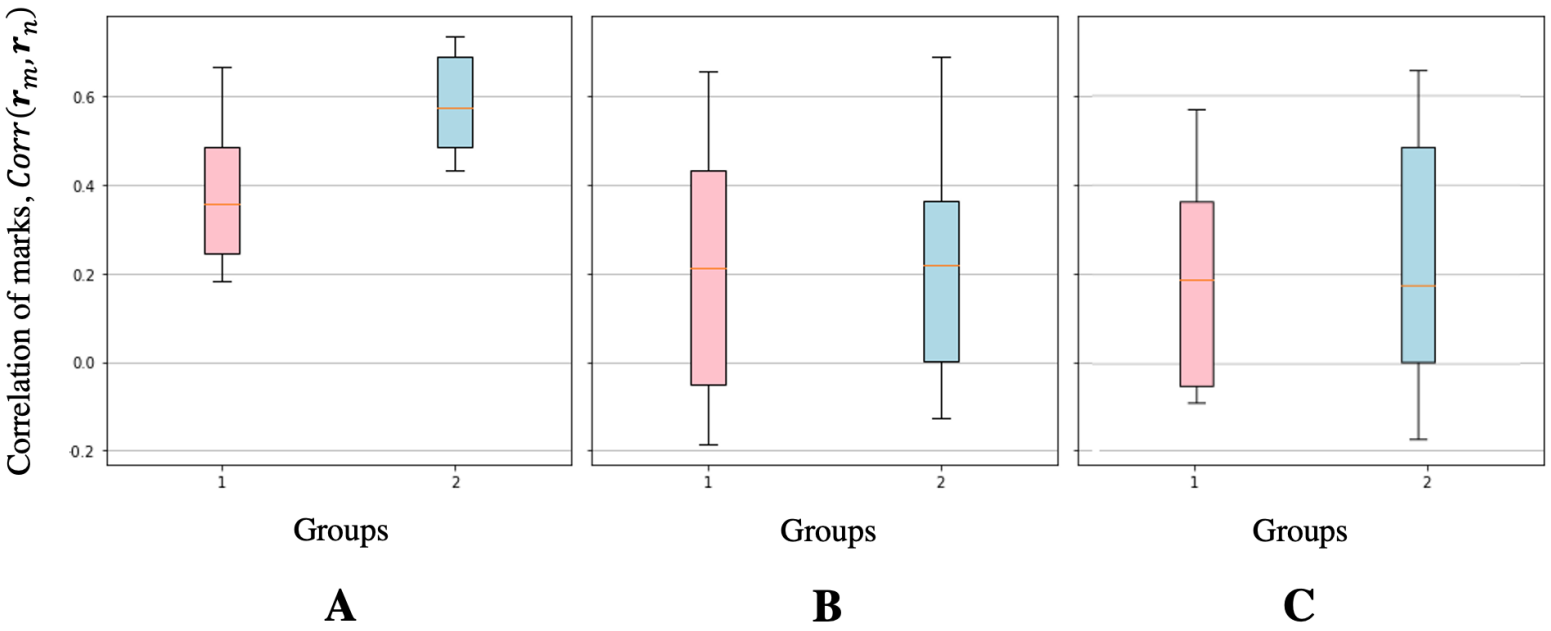

Figure 4: Boxplots showing the correlation between course marks, $Corr(\mathbf{r}_m, \mathbf{r}_n)$, for all pairs of students m and n within each group. A) Minimising the objective gives groups of students with similar skill-sets. B) Maximising the objective gives skill-diverse groups of students. C) Maximising the objective and introducing a lower bound on the balance of a sensitive attribute gives skill-diverse and fair groups of students.

Summary

We have proposed

1. To learn student skill-sets from only course mark data in an unsupervised manner using the graph Laplacian.
2. An algorithm for fair and skill-diverse student group formation.

Finding the optimal solution with graph partitioning is an NP-hard problem. To scale to larger datasets we need heuristic solvers.

We have deployed our algorithm to assign undergraduate students into tutorial groups this academic year in the EEE department. In future work, we will detail our findings.

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UK Research and Innovation

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